

Cloudy Day Report

Group 21

Evaluating Target Group 9

June 3, 2026

Diogo Sousa

Documentation Feedback

- **Style Discrepancy:** The query parameter for pagination size is documented as `pageSize` (camelCase) while the response object uses `page_size` (snake_case). The API Gateway correctly accepts the `pageSize` parameter and returns the corresponding schema, but this inconsistency should ideally be unified.
- **Error Response Documentation:** While standard HTTP error codes (400, 401, 404, 500, 502) are consistently mapped across endpoints using reusable components (e.g., `BadRequestText`), they could be more descriptive. The schema indicates they return a `text/plain` string without explaining *what* triggers them (e.g., what specific validation failure causes a 400, or under what conditions a 404 is thrown for a specific endpoint).

Testing Results

Smoke Testing (Endpoint: `/api/listings/search`)

The smoke test was executed using a Python script checking the endpoint. The test validated the HTTP 200 response, JSON validity, and the presence of all required top-level fields (`total`, `page`, `page_size`, `listings`) and listing level attributes.

- **Basic Paginated Search:** Passed. Returned 5 listings with a reported total of 50,000 records. Latency: $\approx$ 141ms.
- **Filtered Search (Toyota Corolla):** Passed. Returned 5 listings with a reported total of 300 records. Latency: $\approx$ 191ms.

No broken links or malfunctioning behaviors were detected.

Stress Testing (Endpoint: `/api/listings`)

The stress test was executed using Locust against endpoint.

- **Concurrency Settings:** 100 concurrent users, spawning at a rate of 20 users per second, run duration of 30 seconds.
- **Performance observed:**
 - **Total Requests:** 6,899
 - **Failures:** 0 (0.00%)

- **Requests Per Second (RPS):** Peaked at ≈ 275 req/s (average 248.81 req/s over the duration).
- **Latency Statistics:** Average = 72ms, Median = 62ms, 95th percentile = 93ms, 99th percentile = 150ms.

The endpoint proved to be highly stable under the tested load.

Bruno Faustino

Documentation Feedback

[Insert your feedback regarding the clarity, completeness, and structure of the OpenAPI specification provided by the target group.]

Testing Results

Smoke Testing (Endpoint: /example)

[Insert the details of your smoke tests. Describe the tools used, the scenarios evaluated, and whether any broken links or malfunctioning behaviors were found.]

Stress Testing (Endpoint: /example)

[Insert the details of your stress tests. Describe the tools used (e.g., Locust), the concurrency settings, and the observed Requests Per Second (RPS) limit.]

Rodrigo Neto

Documentation Feedback

- **Missing Content-Type Header:** Successful JSON responses do not include the `Content-Type: application/json` header, which can break strict HTTP clients.
- **Poor Overload Behaviour:** Under high load, the backend does not gracefully degrade (e.g., return 429 or 503). Instead, the gateway returns 502 `Bad Gateway` after extremely long timeouts (up to 47 seconds).

Testing Results

Smoke Testing (Endpoint: /api/market/price-comparison)

The smoke test was executed using Postman checking the endpoint. The test validated HTTP 200 responses, JSON validity, pagination, grouping, and sorting behaviors.

- **Happy Path:** Passed with warnings. Valid JSON, correct pagination and sorting. `Content-Type` assertion failed (header omitted).

- **Validation:** Partial pass (3/6). Missing parameters returned 400. Invalid enum values returned 502 **Bad Gateway** instead of 400.

Some malfunctioning behaviors were detected during validation tests.

Stress Testing (Endpoint: `/api/listings/stats/by-location`)

The stress test was executed using Locust against the endpoint.

- **Concurrency Settings:** Step-load ramping from 10 to 200 concurrent users.
- **Performance observed:**
 - **Total Requests:** 3,524
 - **Failures:** 123 (3.49%)
 - **Requests Per Second (RPS):** Peaked at ≈ 50 req/s.
 - **Latency Statistics:** Average = 7,280ms, Median = 460ms, 95th percentile = 34,000ms, 99th percentile = 38,000ms.

The endpoint proved unable to sustain loads above ≈ 50 RPS, leaving the upstream backend in an unhealthy state.
